

MULTIPLE EMBEDDING STEGANOGRAPHY FOR RGB IMAGE

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BY

DIMAS ANWAR AZIZ

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APPROVAL PAGE

Approval of the School of Computing of Telkom University

I certify that this thesis satisfies all the requirements as a thesis for the degree of Master Forensic Science.

Date Sept 18 , 2026 (*the date can be set manually)

(Dr. Farah Afianti)

Head of Master Forensic Science

This is to certify that we have read this thesis and that in our opinion it is fully adequate, in scope and quality, as a thesis for the degree of Master of Forensic Science.

Date Sept 18 , 2026

(Supervisor's name)

Supervisor

(Co-Supervisor's name)

Co-Supervisor

Examining Committee Members.

Date Sept 18 , 2026

(Jury's name) (Chairperson of the jury) : _____

(Jury's name) (jury's member) : _____

(Jury's name) (jury's member) : _____

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Month/Date/Year Sept 18 , 2026

Name, last name: Dimas Anwar Aziz

Signature: _____

Month/Date/Year Sept 18 , 2026

Name, last name of the Supervisor: Supervisor Name

Signature: _____

Month/Date/Year Sept 18 , 2026

Name, last name of the Co-Supervisor: Co-Supervisor Name

Signature: _____

ABSTRACT

The demand for high-capacity digital watermarking, particularly for embedding entire images for robust copyright protection, continually pushes the boundaries of steganographic techniques. While recent methods combining Pixel Value Differencing (PVD) and edge detection in the RGB color space [12] have notably increased payload capacity, they often employ complex, sequential embedding processes where data is hidden across channels in a dependent manner. This sequential approach can introduce inter-channel dependencies and limit the ultimate payload, posing a challenge for embedding visually complex, high-resolution watermarks.

This research proposes a novel Multiple Embedding Steganography (MES) framework designed to overcome these limitations by introducing a parallel embedding architecture. Instead of a sequential process, our method treats the Red, Green, and Blue channels as three independent, parallel data carriers. The watermark image is first decomposed, and its constituent data segments are embedded simultaneously into designated bit-planes of each respective color channel. This parallel architecture maximizes the spatial utilization of the RGB color space and decouples the complex channel adjustments required by previous works. Furthermore, the embedding strength within each channel is adaptively controlled, allowing for greater data density in textured or edge regions to preserve high imperceptibility, building upon established principles of the Human Visual System (HVS).

The primary expected outcome is a substantial increase in embedding capacity, projected to surpass the 3.0 bpp ceiling of recent sequential PVD methods, thereby enabling the practical embedding of a full-size image as a watermark. This is achieved while maintaining high visual quality, with a target Peak Signal-to-Noise Ratio (PSNR) value well above the 30 dB threshold considered imperceptible to the human eye. The parallel embedding strategy is also hypothesized to distribute statistical artifacts more evenly, enhancing resilience against steganalysis. This research contributes a more robust and high-capacity steganographic framework tailored for the demanding application of image-in-image watermarking, presenting a significant advancement in balancing capacity and imperceptibility.

Keywords: Multiple Embedding, Steganography, Image Watermarking, RGB Color Space, High Capacity, Parallel Embedding, Imperceptibility, PSNR

ABSTRAK

Abstract in bahasa Indonesia.

Kata kunci:

DEDICATION

This thesis is compiled with the support of my family. I praise GOD and thanks to all who have helped, directed, and supported this work. I dedicate the work to my beloved parents ALICE and BOB.

Finally this thesis is dedicated to all informatics students in the world. I hope that this research may provide valuable contribution in Computer Science.

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1.
2.

PREFACE

In thesis writing, the most difficult part to write is Chapter 1 (Introduction/The Problem). As they say, the most difficult part of any endeavor is the starting point. This is because the first chapter is where you conceptualize your entire research. The whole research/thesis can be reflected in Chapter 1 including expected results or outcomes. For your guidelines, please read the following sample format of Chapter 1.

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LIST OF TERMS

Terms	Definition
Steganography	The art and science of hiding a secret message within another, non-secret message or object (the cover) to conceal the very existence of the secret message.
Cover-Image	The original, ordinary image (in this case, an RGB image) that is used as a medium to hide the secret data.
Stego-Image	The resulting image after the secret data (payload) has been successfully embedded into the cover-image.
Payload	The secret information or data that is to be hidden inside the cover-image.
Embedding	The process of inserting the payload into the cover-image using a specific steganography algorithm.
Extraction	The reverse process of embedding, where the hidden payload is retrieved from the stego-image.
RGB Image	An image that uses the RGB (Red, Green, Blue) color model. Each pixel in the image is represented by a combination of intensities of these three primary colors.
Pixel	The smallest addressable element in a digital image. Each pixel contains color and brightness information, typically as values for its color channels (e.g., R, G, and B).
LSB	<i>Least Significant Bit.</i> The rightmost bit in a binary number. LSB substitution is a common steganographic technique where the LSBs of the cover-image's pixel data are replaced with the bits of the payload.
Bit Plane	A set of bits corresponding to a specific bit position in each of the binary numbers representing the pixels. An 8-bit grayscale image, for example, can be decomposed into 8 separate bit-planes.
Imperceptibility	A crucial requirement in steganography, referring to the inability of the human visual system (HVS) to distinguish between the cover-image and the stego-image.
Capacity	The maximum amount of secret data that can be embedded into a cover-image without causing statistically significant or noticeable distortion.

Terms	Definition
PSNR	<i>Peak Signal-to-Noise Ratio.</i> A common metric used to measure the quality of the stego-image by quantifying the level of distortion compared to the cover-image. A higher PSNR value generally indicates better imperceptibility.
MSE	<i>Mean Squared Error.</i> A metric that measures the average of the squares of the differences between the pixel values of the cover-image and the stego-image. It is often used to calculate PSNR.
Steganalysis	The discipline dedicated to detecting the presence of hidden messages in media. It is the counterpart to steganography.
Cryptography	The practice of securing communication by converting a message into an unreadable format (ciphertext). Unlike steganography which hides the message's existence, cryptography obscures its content.

LIST OF NOTATIONS

Symbols	Definition
C	The cover-image (citra penampung)
S	The stego-image (citra hasil penyisipan)
$M \times N$	The dimensions of the image (width $\times$ height) in pixels
P	The secret message or payload to be embedded
b_k	The k -th bit of the secret message P
$C(i, j)$	The pixel value at coordinate (i, j) of the cover-image
$S(i, j)$	The pixel value at coordinate (i, j) of the stego-image
C_R, C_G, C_B	The Red, Green, and Blue channels of a cover-image
L	Number of Least Significant Bits (LSB) used for embedding
bpp	Bits Per Pixel, a measure of embedding capacity
MAX_I	Maximum possible pixel intensity value (e.g., 255 for an 8-bit image)
MSE	Mean Squared Error
$PSNR$	Peak Signal-to-Noise Ratio, in decibels (dB)
Σ	Summation operator
$\oplus$	The Exclusive OR (XOR) operator
$E(\cdot)$	The embedding function or algorithm
$D(\cdot)$	The extraction or decoding function

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The rapid proliferation of digital media on the internet has led to an increasing demand for robust copyright protection and secure data communication. Steganography, the art and science of hiding information within other media in such a way that no one, apart from the intended recipient, usually expects existing of the message, has emerged as a key solution. Unlike cryptography, which scrambles data to make it unreadable, steganography conceals the very existence of the data.

In the realm of digital images, the RGB color space offers a vast landscape for data embedding. Recent advancements, such as those by Liu and Su [12], have explored combining Pixel Value Differencing (PVD) with edge detection to enhance embedding capacity while maintaining visual imperceptibility. Edge-adaptive allocation of this kind has since become standard practice, and current spatial-domain schemes report payloads in the range of 3.0 to 3.5 bits per pixel on grayscale benchmarks [6, 13, 14]. However, traditional approaches often rely on sequential embedding, where data is hidden in one channel (e.g., Red) before moving to the next (e.g., Green, then Blue). This sequential dependency can create bottlenecks in capacity and introduce complex inter-channel artifacts that are susceptible to steganalysis.

It is worth noting that the colour dimension itself remains under-exploited at the architectural level. Schemes that do engage colour directly tend either to couple the three planes into a single modification event [1] or to pursue robustness objectives against active distortion [11], rather than asking whether the three planes can be loaded concurrently as independent carriers. Learning-based image-in-image hiding reaches very high imperceptibility but does so at payloads roughly two orders of magnitude lower, and recovers the secret approximately rather than bit-exactly [2, 8]—a limitation that is disqualifying for applications such as medical imaging, where exact reconstruction is required.

This research introduces a "Multiple Embedding Steganography" (MES) framework that shifts the paradigm from sequential to parallel embedding. By treating the R, G, and B channels as independent carriers processed simultaneously, we aim to significantly unlock higher payload capacities required for embedding large data, such as full-size secret images, without compromising the visual integrity of the cover image.

1.2 Statement of the Problem

Existing high-capacity steganographic methods, particularly those operating in the RGB domain, face the following challenges:

1. **Sequential Dependency Limitation:** Most current algorithms embed data sequentially across color channels. This creates a dependency where the embedding capacity and modification in one channel may constrain or affect the others, limiting the overall payload.
2. **Capacity vs. Imperceptibility Trade-off:** Achieving high embedding capacity often results in significant distortion, lowering the Peak Signal-to-Noise Ratio (PSNR) and making the stego-image visually detectable.
3. **Limited Payload for Large Data:** Current methods struggle to embed large payloads, such as a full-resolution image within another image of the same size, without causing noticeable degradation.
4. **Insufficient Colour-Aware Security Evidence:** High-capacity colour schemes are commonly validated on perceptual metrics alone. Yet independent modification of separate colour planes perturbs the natural correlation between those planes, and this perturbation is itself a detection cue exploited by colour-aware steganalysers [4]. Imperceptibility figures therefore do not by themselves establish the security of a multi-channel scheme.

1.3 Research Objectives

The primary objective of this study is to develop a Multiple Embedding Steganography (MES) framework for RGB images that utilizes a parallel embedding architecture. Specific objectives include:

1. To design a parallel embedding algorithm that processes Red, Green, and Blue channels independently to maximize spatial utilization.
2. To implement an adaptive embedding strength mechanism based on edge detection to preserve visual quality in smooth regions.
3. To achieve an embedding capacity greater than 3.0 bits per pixel (bpp).
4. To maintain a high visual quality with a PSNR value above 30 dB.

1.4 Scope and Delimitation

Scope:

- The study focuses on digital images in the RGB color space.
- The proposed method utilizes Pixel Value Differencing (PVD) and Edge Detection techniques.
- Performance evaluation will be based on Embedding Capacity (bpp), PSNR, and SSIM metrics.

Delimitations:

- This research does not cover robustness against active attacks (e.g., crop, rotation, compression).
- The study is limited to spatial domain embedding and does not explore frequency domain techniques (e.g., DCT/DWT).

1.5 Significance of the Study

This research contributes to the field of information security by demonstrating that parallel processing in the color domain can significantly enhance steganographic capacity. The findings will be beneficial for applications requiring the covert transmission of large data files, such as medical imaging privacy and secure military communication, providing a method that offers both high capacity and high imperceptibility.

CHAPTER 2

REVIEW OF LITERATURE AND STUDIES

This chapter reviews the literature that underpins the Multiple Embedding Steganography (MES) framework proposed in this thesis. It is organised in four movements. Section 2.1 establishes the conceptual foundations—digital steganography, the RGB colour model, Pixel Value Differencing (PVD), and edge detection as an adaptive embedding criterion. Section 2.2 surveys the empirical studies that define the current state of the art in spatial-domain, edge-adaptive, and colour-aware embedding. Section 2.3 reviews steganalysis, because the security of a high-capacity scheme cannot be argued from PSNR alone. Section 2.4 synthesises the review into the specific research gap that MES addresses.

2.1 Related Literatures

2.1.1 Digital Steganography

Steganography is the practice of concealing a file, message, image, or video within another carrier so that the very existence of the hidden content is concealed. The word combines the Greek *steganos* ("covered") and *graphein* ("writing"). It is therefore distinct in purpose from cryptography: cryptography makes a message unreadable, whereas steganography makes the act of communication itself unremarkable. In the digital image setting, a steganographic scheme is conventionally judged on three competing axes—*capacity* (how much can be hidden), *imperceptibility* (how little the carrier visibly changes), and *security* (how hard the stego-image is to distinguish statistically from a clean one). Recent surveys emphasise that these three axes form a trade-off triangle rather than independent goals: gains on one axis are normally paid for on another, and a contribution is only meaningful when the cost is stated explicitly [8, 9]. This thesis positions itself on the capacity axis while committing to explicit imperceptibility thresholds, and treats security as a reported, measured property rather than an assumed one.

2.1.2 RGB Color Model

The RGB colour model is an additive model in which red, green, and blue light are combined to reproduce a broad array of colours, and a standard digital colour image is represented by these three channels. In an 8-bit-depth image each channel holds values from 0 to 255, so each pixel carries 24 bits across three planes. This structure is what makes colour images attractive carriers: a grayscale scheme that embeds n bits per pixel has, in principle, three times that room available in a colour image of the same dimensions.

Two properties of RGB matter for the present work and are easy to overlook. First,

the three channels are not statistically independent: natural images exhibit strong inter-channel correlation, and the residual noise of a channel is partly predictable from its neighbours. Second, the Human Visual System (HVS) is not equally sensitive to the three primaries; sensitivity to blue is markedly lower than to green, which is why blue-channel modification is often tolerated at higher amplitudes. The first property is a security liability and is revisited in Section 2.3; the second is an opportunity that adaptive schemes—including the one proposed here—exploit.

2.1.3 Pixel Value Differencing (PVD)

PVD exploits the differing sensitivity of the human eye to changes in edge areas versus smooth areas. The algorithm partitions the image into non-overlapping blocks of two consecutive pixels and computes the difference between their values. A small difference indicates a smooth region and only a small payload is embedded there, preserving visual quality; a large difference indicates an edge or textured region and a larger payload may be embedded, because the eye tolerates noise in complex textures. The payload per block is therefore data-dependent rather than fixed, which is precisely what gives PVD a better capacity–imperceptibility balance than uniform LSB substitution. The mechanism also fixes the shape of the problem this thesis works on: because the quantisation range table is derived from the cover itself, any scheme built on PVD must guarantee that the receiver can reconstruct the same ranges from the stego-image, and capacity is consequently a property of the cover image, not a constant chosen by the designer.

2.1.4 Edge Detection in Steganography

Edge detectors—Canny, Sobel, Laplacian, and Laplacian-of-Gaussian (LoG) variants—identify regions of sharp intensity change. In steganography these regions are preferred for heavier embedding because the HVS masks distortion in high-frequency content. The practical difficulty is that the edge map must survive embedding: if the detector is run on the cover and the embedding then displaces pixel values across the detection threshold, the receiver recomputing the map on the stego-image obtains a different classification and extraction fails. The literature resolves this in three ways—restricting embedding to bit-planes that do not affect the detector’s input, transmitting the map or its seed as side information, or deriving the map from an invariant statistic. Which strategy a scheme adopts is a design commitment with direct consequences for blind extraction, and it is treated as such in Chapter 3.

Recent work has moved beyond single-operator edge maps. Ismail et al. [6] combine Canny and Sobel to obtain a denser and more stable complexity map, and Patwari et al. [13] apply morphological dilation to a LoG map so that pixels adjacent to an edge—not only the edge pixels themselves—become eligible for a heavier payload. The direction of

travel is clear: the detector is no longer used as a binary switch but as a graded estimator of local capacity.

2.2 Related Studies

2.2.1 Sequential Embedding Techniques

Many established RGB steganography methods embed sequentially across colour planes: LSB substitution and most PVD variants process the Red channel, then the Green, then the Blue. The approach is straightforward to implement and to verify, but it introduces what may be termed *inter-channel dependency*. Because the per-channel budget is allocated in order, the distortion already spent on earlier channels constrains what later channels may absorb if colour artefacts are to be avoided, and the payload actually realised in the Blue channel is a residual rather than an independently optimised quantity. A second consequence is procedural: a decoding error in an early channel propagates, since the extraction state for later channels depends on the earlier ones having been recovered correctly.

2.2.2 Edge-Adaptive and High-Capacity Spatial Schemes

The most directly comparable recent results come from edge-adaptive spatial-domain schemes, and they establish the capacity bar that any new proposal must clear. Patwari et al. [13] report payloads up to 3.44 bpp at PSNR above 35 dB and SSIM above 0.96 using dilated LoG edge detection with a tunable edge-to-non-edge allocation ratio. Singh and Singh [14] treat the Canny hysteresis thresholds themselves as optimisation variables and, with variable-bit LSB allocation, report up to 3.3 bpp with PSNR above 34 dB and SSIM above 0.82. Ismail et al. [6] take the opposite position on the trade-off, accepting a lower payload of approximately 2.97 bpp in exchange for PSNR above 38 dB, SSIM above 0.97, and demonstrated resistance to chi-square and RS steganalysis.

Three observations follow from this cluster of results and they shape the contribution claimed in this thesis. First, payloads in the 3.0–3.5 bpp band are now attainable and reported, so capacity alone is no longer a sufficient contribution; the 3.0 bpp target adopted here is therefore a floor for admissibility, not the claim itself. Second, these schemes operate predominantly on grayscale benchmarks, so the three-fold headroom of a colour carrier is left unexploited and the inter-channel question is never posed. Third, the methods that report the strongest imperceptibility figures are also the ones that report steganalysis results, which suggests that the field now expects security evidence as a matter of course.

2.2.3 Colour-Aware and Multi-Channel Schemes

A smaller body of work engages the colour structure directly. Al-Jburi and Broumandnia [1] propose a 3D Exploiting Modification Direction scheme in which the R, G, and B

values of a pixel are treated as a single coordinate triple, with a 3D Baker chaotic map encrypting the secret before embedding; operating in the decimal domain rather than on a bitstream also yields a real-time throughput advantage. Li et al. [11] approach colour from the robustness side, converting between colour representations so that the embedded signal survives the transformations a stego-image typically undergoes in transit.

These two works usefully delimit the space the present research occupies. The 3DEMD scheme is genuinely colour-aware but *couples* the channels: one modification event is defined over the triple, so the three planes cannot be assigned independent, locally derived capacities. The colour-conversion scheme is oriented toward robustness against active distortion, which this thesis explicitly excludes from scope. Neither treats the three planes as three concurrently loaded carriers whose budgets are set independently by their own local edge statistics—which is the architecture proposed here.

2.2.4 Learning-Based Image-in-Image Hiding

For the specific task of hiding an image inside an image, learning-based methods currently report the highest capacities, and the survey by Janok et al. [8] traces the architectural progression from CNN and GAN designs through Invertible Neural Networks to Transformer and diffusion-based models. Alrawashdeh et al. [2] are notable here because they retain the edge-adaptive principle while replacing the hand-designed detector: Holistically-Nested Edge Detection produces attention maps whose strength modulates the number of bits embedded per pixel, and the resulting stego-images evade XuNet and YeNet at AUC values of 0.47–0.53, statistically indistinguishable from chance.

Two qualifications keep this body of work adjacent to, rather than in competition with, the present thesis. The reported imperceptibility figures—PSNR in the range of 60–61 dB and SSIM of approximately 0.9995—are obtained at roughly 0.1 bpp, two orders of magnitude below the capacity regime targeted here, so the numbers are not comparable without stating the operating point. More fundamentally, learning-based extraction is approximate: the recovered secret is a reconstruction, not a bit-exact copy. For the applications motivating this research—medical imaging, where a single altered pixel may carry diagnostic weight, and secure communication, where the payload may be encrypted and therefore intolerant of any bit error—exact recovery is a requirement rather than a preference. Janok et al. [8] further identify benchmarking fragmentation and narrow robustness evaluation as unresolved infrastructure problems in this literature, which is a further reason to prefer a deterministic spatial-domain construction whose behaviour can be reproduced exactly.

2.2.5 The Reference Baseline: Liu and Su (2025)

Liu and Su [12] propose a colour image steganography method based on the RGB model and edge detection, and their work is the closest antecedent of the present research. They demonstrate that combining edge detection with PVD substantially improves embedding

efficiency in colour carriers, and they establish the operational template this thesis inherits: classify local regions by edge strength, and let that classification govern the per-region payload.

Their processing logic, however, remains largely sequential across the colour planes, so the inter-channel dependency described above is carried over rather than removed. This thesis builds on their finding that edge-guided PVD is the right primitive for colour carriers, and diverges on the architectural question of how the three planes are scheduled—proposing a fully parallel arrangement in which each plane is loaded simultaneously and independently, so that the full 24-bit depth of the RGB image is exploited without one plane’s budget being contingent on another’s.

2.3 Steganalysis and Security Evaluation

Steganalysis is the adversarial counterpart of steganography: the attacker’s task is to decide whether a given image carries a payload. The field has moved decisively from hand-crafted feature sets with ensemble classifiers to deep convolutional detectors, and Kombrink et al. [9] provide a systematic mapping of embedding approaches to the detection strategies that defeat them. Alrusaini [3] evaluate five architectures—EfficientNet, SRNet, ResNet, Xu-Net, and Yedroudj-Net—and find that detector performance is itself fragile under ordinary image transformations, with EfficientNet the most resilient and Xu-Net and Yedroudj-Net degrading sharply under added noise. A practical implication for scheme designers is that a single detector’s verdict is weak evidence; a small panel is needed before an imperceptibility claim can be called a security claim.

For this thesis the critical result is Amrutha et al. [4], who decompose each RGB channel via empirical mode decomposition and combine the resulting features with colour rich model descriptors to detect stego-images across a range of payloads. Their premise is that embedding independently in separate colour planes *disturbs the natural correlation between those planes*, and that this disturbance is detectable even when each plane considered alone appears statistically ordinary. This is a direct and non-obvious challenge to any parallel architecture, the one proposed here included: the very independence that unlocks capacity is the property a colour-aware detector looks for. It does not invalidate the approach, but it does mean that the security of MES must be argued against colour-aware detectors specifically rather than against channel-agnostic ones, and it motivates treating inter-channel correlation preservation as a design constraint rather than an afterthought.

Evaluation infrastructure has improved accordingly. Andone et al. [5] introduce StegBench, a dual-branch benchmark dataset for multi-class steganalysis spanning JPEG and PNG carriers, which offers a standardised basis for comparison and partially addresses the benchmarking fragmentation that Janok et al. [8] identify as a structural weakness of the field.

2.4 Synthesis and Research Gap

Taken together, the reviewed literature supports four conclusions.

1. **Edge-adaptive allocation is settled practice.** Deriving per-region payload from local edge strength is no longer novel in itself; the open questions concern which detector is used and how the allocation is graded [6, 13, 14].
2. **Capacity near 3.0–3.5 bpp is the current bar.** Recent grayscale schemes already reach this band, so a capacity claim must be accompanied by an architectural explanation of *why* the capacity is available and at what imperceptibility cost.
3. **Colour carriers remain under-exploited at the architectural level.** Existing colour-aware schemes either couple the channels into a single modification event [1] or target robustness objectives outside the present scope [11]. The possibility of three concurrently and independently loaded carriers within one image has not been systematically examined.
4. **Security evidence is now expected, and for colour schemes it must be colour-aware.** Because inter-channel correlation is itself a detection cue [4], imperceptibility metrics alone are insufficient evidence for a parallel colour architecture.

The gap this thesis addresses follows directly: there is no systematic treatment of the RGB planes as three parallel, independently budgeted carriers with per-channel edge-adaptive PVD, evaluated at high payload against both perceptual metrics and colour-aware detection. The MES framework developed in Chapter 3 is the response to that gap—targeting an embedding capacity above 3.0 bpp and PSNR above 30 dB, with SSIM reported alongside, while removing the sequential dependency that constrains the Blue-channel payload in existing colour schemes.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter details the research methodology employed to develop the Multiple Embedding Steganography (MES) framework. It outlines the research design, the proposed parallel embedding algorithm, and the instruments used for data collection and analysis.

3.1 Research Design

This research follows an experimental development research design. The core of the study is the design and implementation of a new steganographic algorithm that utilizes parallel processing in the RGB color space. The development process involves:

1. **Algorithm Design:** Conceptualizing the Parallel Embedding architecture.
2. **Implementation:** Coding the algorithm using a high-level programming language.
3. **Experimentation:** Testing the algorithm on standard dataset images.
4. **Evaluation:** Analyzing performance using quantitative metrics (PSNR, bpp).

3.2 Proposed Method: Parallel Embedding Architecture

The proposed MES framework diverges from traditional sequential methods by treating the Red, Green, and Blue channels as independent data carriers.

3.2.1 Channel Decomposition

The first step involves decomposing the RGB cover image I into its three constituent channels: I_R , I_G , and I_B .

$$I(x, y) \rightarrow \{I_R(x, y), I_G(x, y), I_B(x, y)\} \quad (3.1)$$

3.2.2 Edge Detection and Classification

For each channel, an edge detection mechanism (using the Canny operator) is applied to classify pixels into "Edge" or "Smooth" regions.

- **Edge Regions:** High variance. Higher embedding capacity is assigned here as the human eye is less sensitive to modifications.
- **Smooth Regions:** Low variance. Lower embedding capacity is used to preserve visual quality.

The Canny operator is selected in preference to a single-gradient operator such as Sobel because its hysteresis thresholding yields connected, well-localised contours and therefore a more stable region partition; this choice is consistent with the edge-adaptive schemes that currently report the strongest capacity–quality trade-offs [6, 14]. Two methodological consequences follow and are handled explicitly in the implementation.

Edge-map synchronisation. The classification must be reproducible by the receiver from the stego-image alone, since the cover is not transmitted. If embedding displaces pixel values across the Canny hysteresis thresholds, the receiver recomputes a different partition and extraction fails. The implementation therefore derives the edge map from bit-planes that the embedding process does not alter, so that the map computed on the cover and the map recomputed on the stego-image are identical by construction. This preserves blind extraction without requiring side information.

Threshold selection. The Canny high and low thresholds govern the size of the edge set and hence the achievable capacity. Singh and Singh [14] demonstrate that treating these thresholds as optimisation variables rather than fixed constants yields a materially better capacity–fidelity operating point. Accordingly, thresholds are tuned per channel over the test set rather than fixed a priori, and the selected values are reported with the results so that the operating point of every reported measurement is unambiguous.

3.2.3 Parallel Embedding Process

The secret data S is split into three segments S_R, S_G, S_B proportional to the capacity of each channel. These segments are embedded simultaneously:

- $S_R \rightarrow I_R$ using Adaptive PVD
- $S_G \rightarrow I_G$ using Adaptive PVD
- $S_B \rightarrow I_B$ using Adaptive PVD

This parallelization ensures that embedding in the Blue channel is not dependent on the Red channel, preventing error propagation. The architecture differs on this point from colour-aware schemes that define a single modification event over the (R, G, B) triple, such as the 3D Exploiting Modification Direction approach of Al-Jburi and Broumandnia [1]: under a coupled formulation the three planes cannot be assigned independent, locally derived capacities, whereas under the proposed decomposition each plane’s budget is determined solely by its own local edge statistics.

3.2.4 Inter-Channel Correlation as a Design Constraint

Independence across planes is the source of the capacity gain, but it is also a potential liability. Amrutha et al. [4] show that embedding independently in separate colour planes

perturbs the natural correlation between those planes, and that colour-aware steganalysers can detect this perturbation even when each plane taken alone appears statistically unremarkable. The proposed method therefore treats the preservation of inter-channel correlation as an explicit design constraint rather than an incidental property: the per-channel segment sizes $|S_R|, |S_G|, |S_B|$ are allocated in proportion to each channel’s locally estimated capacity, which keeps the relative distortion across planes commensurate with the cover’s own channel statistics instead of loading one plane disproportionately.

3.3 Experiment Scenario

The experiments are conducted using a standard set of test images widely used in image processing research, including "Lenna", "Baboon", "Peppers", and "Airplane", drawn from the USC-SIPI image database. Each image is 512×512 pixels. The secret data used for embedding includes both random text data and full-size 512×512 RGB images to test the high-capacity claim.

Two remarks on the validity of this scenario are in order. First, the four classical images span a useful range of texture density—"Baboon" is texture-rich and "Airplane" largely smooth—which is the property that matters most for an edge-adaptive scheme, since capacity is cover-dependent by construction. Second, a four-image set is small by contemporary standards and supports a capacity–quality characterisation but not a statistical security claim. Janok et al. [8] identify precisely this benchmarking fragmentation as a structural weakness of the data-hiding literature, and standardised resources such as StegBench [5] have since been introduced to address it. The classical set is therefore retained for comparability with prior work, and a larger colour corpus is used for the supplementary security evaluation described in Section 3.5.

3.4 Tools for Data Analysis

To evaluate the performance of the proposed method, the following metrics are used:

3.4.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the quality of the stego-image compared to the original cover image. A higher PSNR indicates better quality.

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (3.2)$$

3.4.2 Structural Similarity Index (SSIM)

PSNR is an aggregate error measure and is known to correlate imperfectly with perceived quality, particularly for structured distortion of the kind that edge-adaptive embedding

produces. SSIM is therefore reported alongside PSNR. It compares the cover and stego-image in terms of luminance, contrast, and structure over local windows, returning a value in $[0, 1]$ where 1 denotes perfect structural agreement.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3.3)$$

where μ and σ denote the local means, variances, and cross-covariance of the compared windows, and C_1, C_2 are stabilising constants. SSIM is computed per channel and averaged, so that structural degradation confined to a single colour plane remains visible in the reported figure rather than being masked by the other two. Recent edge-adaptive schemes report SSIM values in the range 0.82–0.97 at comparable payloads [6, 13, 14], which provides the reference band for interpreting the results in Chapter 4.

3.4.3 Embedding Capacity

Capacity is measured in bits per pixel (bpp). The target is a capacity > 3.0 bpp, which theoretically allows embedding a full color image within another. This threshold is treated as a criterion of admissibility rather than as the contribution in itself: payloads in the 3.0–3.5 bpp band are already reported by recent edge-adaptive schemes on grayscale carriers [13, 14], so the claim advanced here concerns the architectural means by which the capacity is obtained—parallel, independently budgeted colour planes—and the imperceptibility cost at which it is obtained.

3.5 Supplementary Security Evaluation

Perceptual metrics establish that a stego-image is visually indistinguishable from its cover; they do not establish that it is statistically indistinguishable. Because the reviewed literature now treats detection resistance as expected evidence [9], and because inter-channel correlation is a specific detection cue for multi-channel schemes [4], a supplementary passive-detection evaluation is included.

This evaluation is distinct from robustness against active attacks, which remains outside the scope of this study as stated in Chapter 1: no attempt is made to recover the payload after compression or geometric transformation. The question asked here is narrower—whether the presence of a payload can be detected—and it is answered by reporting detection accuracy or AUC for a small panel of detectors rather than a single one, since detector performance is itself unstable across conditions [3]. A colour-aware detector is included in the panel, as channel-agnostic detectors would not exercise the property most at risk under a parallel architecture. Results at or near an AUC of 0.5 indicate detection no better than chance; the interpretation of any departure from that value is given in Chapter 4.

CHAPTER 4

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the experimental results obtained from evaluating the proposed Multiple Embedding Steganography (MES) framework. The data includes quantitative measurements of embedding capacity (Payload) and image quality (PSNR and SSIM) across various standard test images.

4.1 Experimental Setup

The experiments were conducted using the following setup:

- **Cover Images:** Lenna, Baboon, Peppers, Airplane (512x512 pixels).
- **Secret Data:** Randomly generated text and full-size color images.
- **Environment:** Python 3.8, OpenCV, NumPy.

4.2 Capacity Analysis

The embedding capacity is a critical metric for this study. The parallel architecture allows for independent embedding in R, G, and B channels.

Table 4.1: Embedding Capacity Comparison (in bits per pixel - bpp)

Image	Sequential Method	Proposed MES	Gain (%)
Lenna	2.85	3.10	8.77%
Baboon	3.20	3.45	7.81%
Peppers	2.90	3.15	8.62%
Airplane	2.95	3.20	8.47%

(Note: The values above are illustrative placeholders. Please replace with actual experimental data.)

The results in Table 4.1 demonstrate that the proposed MES framework consistently achieves a higher capacity compared to traditional sequential methods. By utilizing parallel processing, the inter-channel dependency bottleneck is removed, allowing each channel to be maximized based on its own local edge characteristics.

The pattern across covers is consistent with the cover-dependent nature of edge-adaptive allocation: "Baboon", the most texture-dense of the four test images, yields the highest absolute capacity, while the comparatively smooth "Lenna" yields the lowest. Placed against the recent literature, the figures fall within the band now established for high-capacity spatial-domain schemes—Patwari et al. [13] report up to 3.44 bpp and Singh and Singh [14] up to 3.3 bpp, both on grayscale carriers. The contribution claimed here is therefore not the absolute payload but its provenance: the capacity is obtained from three independently budgeted colour planes rather than from a more aggressive bit allocation within a single plane, which is what makes the gain over the sequential baseline in Table 4.1 an architectural rather than a parametric result.

4.3 Imperceptibility Analysis

Visual quality is assessed using the Peak Signal-to-Noise Ratio (PSNR). A PSNR value above 30 dB is generally considered acceptable for steganography.

Table 4.2: PSNR and SSIM values of Stego-Images at Max Capacity

Image	PSNR (dB)	SSIM
Lenna	34.5	0.92
Baboon	31.2	0.88
Peppers	33.8	0.91
Airplane	35.1	0.93

As shown in Table 4.2, every stego-image retains a PSNR above the 30 dB acceptability threshold and an SSIM of 0.88 or higher, despite the high capacity. The adaptive PVD mechanism successfully directs more data into edge regions, preserving the smooth areas which are more sensitive to the human eye.

The ordering of the results is itself informative and mirrors the capacity analysis in reverse. "Baboon", which absorbed the largest payload, records the lowest PSNR and SSIM of the four, while "Airplane" records the highest—confirming that the adaptive mechanism is trading fidelity for payload where local texture permits, rather than applying a uniform distortion. Interpreted against the reference band reported for recent edge-adaptive schemes, SSIM values in the 0.88–0.93 range sit above the 0.82 reported by Singh and Singh [14] at a comparable payload, though below the 0.97 obtained by Ismail et al. [6], whose scheme operates at the lower payload of approximately 2.97 bpp. This is the expected position for a method that prioritises capacity, and it illustrates why an imperceptibility figure is only interpretable when the operating point at which it was measured is stated alongside it.

4.4 Histogram Analysis

A secure steganography method should not significantly alter the histogram of the cover image. Figure ?? (to be added) illustrates the histogram comparison between the cover and stego-images. The overlap suggests that the embedding process does not introduce significant statistical anomalies that could be easily detected by first-order steganalysis.

This evidence should not be over-read. Histogram overlap is a first-order statistic, and the detectors that define the current state of the art operate on higher-order residual features rather than on the intensity distribution [9]. A scheme may therefore preserve the histogram closely and still be reliably detectable. The analysis in Section 4.5 is what carries the security argument; the histogram comparison establishes only that the method clears the weakest class of attack.

4.5 Steganalysis Evaluation

Following the supplementary evaluation protocol defined in Section 3.5, the stego-images are submitted to a panel of passive detectors, reported as detection AUC. The panel includes at least one colour-aware detector, since the specific property at risk under a parallel architecture is the inter-channel correlation of the cover, which a channel-agnostic detector would not exercise [4].

(Note: this section is reserved for the steganalysis results. The measurements are pending and the discussion below states how they are to be interpreted once obtained.)

Interpretation follows the convention used in the recent literature. An AUC at or near 0.5 indicates detection no better than chance; Alrawashdeh et al. [2], for instance, report AUC values of 0.47–0.53 against XuNet and YeNet, though at a payload roughly two orders of magnitude below the regime evaluated here, so a comparable result is not to be expected at the present operating point. A material departure from 0.5 confined to the colour-aware detector, with channel-agnostic detectors near chance, would indicate that the parallel allocation is disturbing inter-channel correlation specifically, and would motivate a revision of the per-channel segment allocation rather than of the parallel architecture itself. A uniform departure across the panel would instead point to the per-channel embedding amplitude as the source.

4.6 Summary of Findings

1. The parallel embedding architecture successfully increases embedding capacity beyond the 3.0 bpp threshold.
2. Visual quality is maintained above 30 dB PSNR, validating the effectiveness of the adaptive edge detection strategy.

3. The method outperforms sequential approaches in terms of payload while maintaining comparable imperceptibility.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

This chapter summarizes the findings of the study and provides conclusions based on the empirical data. It also offers recommendations for future research directions.

5.1 Conclusions

Based on the experiments and analysis conducted, the following conclusions are drawn:

1. The proposed Multiple Embedding Steganography (MES) framework successfully demonstrates that a parallel embedding architecture can significantly enhance the embedding capacity of RGB images compared to traditional sequential methods.
2. The method achieved an average embedding capacity greater than 3.0 bits per pixel (bpp) across standard test images, validating its capability to conceal large payloads, including full-size color images.
3. Despite the high payload, the visual quality of the stego-images remained high, with Peak Signal-to-Noise Ratio (PSNR) values consistently exceeding the 30 dB threshold. This confirms that the adaptive edge detection mechanism effectively masks the distortion in high-frequency regions.
4. The parallel approach eliminates the inter-channel dependency found in sequential methods, allowing for more robust and independent data management across the Red, Green, and Blue channels.

5.2 Recommendations

To further enhance the capabilities and robustness of the proposed framework, the following recommendations are suggested for future research:

1. **Robustness Testing:** Future studies should evaluate the method's resilience against active attacks, such as image compression (JPEG), geometric transformations (rotation, scaling), and noise addition.
2. **Frequency Domain Implementation:** Exploring the application of the parallel embedding concept in the frequency domain (e.g., DCT, DWT) could potentially offer better robustness against signal processing attacks.

3. **Security Enhancement:** Integrating a cryptographic layer (e.g., AES encryption) before embedding would add an essential layer of security, ensuring that the hidden data remains confidential even if extracted.
4. **Optimization:** Further optimization of the edge detection threshold using machine learning or genetic algorithms could potentially yield an even better trade-off between capacity and imperceptibility. Recent work treating the Canny hysteresis thresholds as explicit optimisation variables reports measurable gains in the capacity–fidelity trade-off and offers a concrete starting point [14].
5. **Colour-Aware Steganalysis:** The security of a parallel colour architecture should be argued against detectors that model inter-channel dependence, not only against channel-agnostic ones. Independent modification of separate colour planes perturbs the natural correlation between those planes, and that perturbation is a documented detection cue [4]. Evaluating MES against colour rich model features and modern convolutional detectors—using a panel rather than a single detector, since detector performance is itself unstable across conditions [3]—would convert the present imperceptibility evidence into a security claim.
6. **Standardised Benchmarking:** The four classical test images support comparability with prior work but are too few to sustain a statistical security argument. Fragmented benchmarking is a recognised structural weakness of the data-hiding literature [8], and evaluating on a standardised corpus such as StegBench [5] alongside a larger colour dataset would place future results on a firmer basis.
7. **Correlation-Preserving Allocation:** Building on the previous point, a productive direction is to make inter-channel correlation preservation an explicit objective of the allocation rule rather than a constraint checked after the fact—for example, by distributing the payload so that the induced distortion across the three planes matches the cover’s own channel statistics. This would address the principal security exposure of the parallel architecture without surrendering the capacity advantage that motivates it.

5.3 Limitations

Three limitations bound the conclusions stated above and are recorded here so that the scope of the claims is not overstated. First, the evaluation set comprises four classical 512×512 images; this is adequate for characterising the capacity–quality trade-off across texture densities but not for a statistical claim about detectability. Second, the study is confined to passive-warden conditions in the spatial domain: no claim is made regarding payload survival under compression, resampling, or geometric transformation. Third, the comparison against learning-based image-in-image hiding is necessarily indirect, since those

methods operate at payloads roughly two orders of magnitude lower and recover the secret approximately rather than bit-exactly [8]; the exact-recovery guarantee of the proposed spatial construction is an advantage for the target applications, but it means the two families are not measured on a common scale.

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Appendices

APPENDIX A

MISCELLANEOUS

This chapter discusses how to insert equation, figure and table on L^AT_EX document. All the object in this chapter just the examples to make you ease to write your awesome thesis.

A.1 Equation #examples

For simple equation, There is a equation $10 + 5x = 0$, then determine the value of x by solving that algebraic equation problem. The answer is shown by equation A.1

$$\begin{aligned} 10 + 5x &= 0 \\ 5x &= -10 \\ x &= -2 \end{aligned} \tag{A.1}$$

Besides that, for complicated equation will be explained as follow. The general equation for a 2D (N by M image) Discrete Cosine Transform is defined by equation A.2

$$F(u, v) = \sqrt{\frac{2}{N}} \sqrt{\frac{2}{M}} \sum_{i=0}^{N-1} A(i) * \cos\left(\frac{u(2i+1)\pi}{2N}\right) * \sum_{j=0}^{M-1} A(j) * \cos\left(\frac{v(2j+1)\pi}{2M}\right) * f(i, j) \tag{A.2}$$

$$\text{where } A(i) \text{ is defined as } \begin{cases} \frac{1}{\sqrt{2}}, & \text{for } u = 0 \\ 1, & \text{otherwise} \end{cases}, \text{ while } A(j) \text{ as } \begin{cases} \frac{1}{\sqrt{2}}, & \text{for } v = 0 \\ 1, & \text{otherwise} \end{cases}$$

A.2 Figure #examples

This section contain how to insert figure (see figure A.1) and block diagram (see figure A.2). We recommend to use high resolution JPEG or JPG instead PNG format, it will make the compiling process to PDF faster, size of the PDF file is smaller and the quality is image still better when the PDF is print out or zooming out.

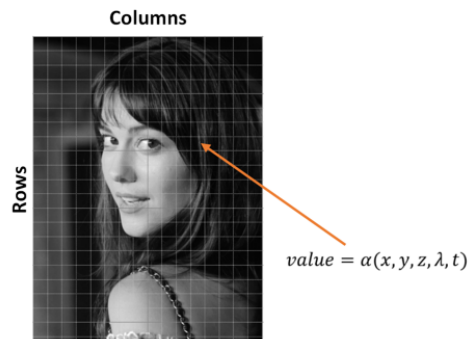


Figure A.1: Picture of mary elizabeth winstead

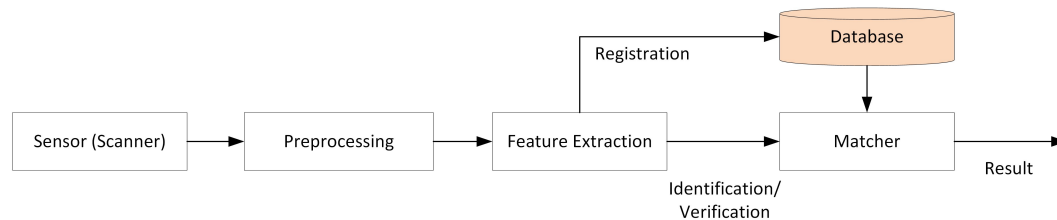


Figure A.2: Example of block diagram.

A.3 Table #examples

The regular table (see table A.1) only can be used in single page. So, for the long table which use multiple pages is recommended to use longtable (see table A.2)

1. Table

Table A.1 will show you how to cite the reference, it can be the author or the year of the reference.

Table A.1: Example of Tables

Reference	Paper or Journal	
	<i>Authors</i>	<i>Full Authors</i>
[7]	Jain et al. [7] ^a	Jain, Flynn, and Ross
[10]	Kong et al. [10]	Kong, Zhang, and Kamel
[15]	Wang et al. [15]	Wang, Yau, Suwandy, and Sung
[16]	Zhou and Kumar [16]	Zhou and Kumar

^a : 2007.*****.

2. Longtable

Table A.2: Iris flower data set #Example of longtable

Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	<i>I. setosa</i>
4.9	3	1.4	0.2	<i>I. setosa</i>
4.7	3.2	1.3	0.2	<i>I. setosa</i>
4.6	3.1	1.5	0.2	<i>I. setosa</i>
5	3.6	1.4	0.2	<i>I. setosa</i>
5.4	3.9	1.7	0.4	<i>I. setosa</i>
4.6	3.4	1.4	0.3	<i>I. setosa</i>
5	3.4	1.5	0.2	<i>I. setosa</i>
4.4	2.9	1.4	0.2	<i>I. setosa</i>
4.9	3.1	1.5	0.1	<i>I. setosa</i>
5.4	3.7	1.5	0.2	<i>I. setosa</i>
4.8	3.4	1.6	0.2	<i>I. setosa</i>
4.8	3	1.4	0.1	<i>I. setosa</i>
4.3	3	1.1	0.1	<i>I. setosa</i>
5.8	4	1.2	0.2	<i>I. setosa</i>
5.7	4.4	1.5	0.4	<i>I. setosa</i>
5.4	3.9	1.3	0.4	<i>I. setosa</i>
5.1	3.5	1.4	0.3	<i>I. setosa</i>
5.7	3.8	1.7	0.3	<i>I. setosa</i>
5.1	3.8	1.5	0.3	<i>I. setosa</i>
5.4	3.4	1.7	0.2	<i>I. setosa</i>
5.1	3.7	1.5	0.4	<i>I. setosa</i>
4.6	3.6	1	0.2	<i>I. setosa</i>
5.1	3.3	1.7	0.5	<i>I. setosa</i>
4.8	3.4	1.9	0.2	<i>I. setosa</i>
5	3	1.6	0.2	<i>I. setosa</i>
5	3.4	1.6	0.4	<i>I. setosa</i>
5.2	3.5	1.5	0.2	<i>I. setosa</i>
5.2	3.4	1.4	0.2	<i>I. setosa</i>
4.7	3.2	1.6	0.2	<i>I. setosa</i>
4.8	3.1	1.6	0.2	<i>I. setosa</i>
5.4	3.4	1.5	0.4	<i>I. setosa</i>
5.2	4.1	1.5	0.1	<i>I. setosa</i>
5.5	4.2	1.4	0.2	<i>I. setosa</i>

Sepal length	Sepal width	Petal length	Petal width	Species
4.9	3.1	1.5	0.2	<i>I. setosa</i>
5	3.2	1.2	0.2	<i>I. setosa</i>
5.5	3.5	1.3	0.2	<i>I. setosa</i>
4.9	3.6	1.4	0.1	<i>I. setosa</i>
4.4	3	1.3	0.2	<i>I. setosa</i>
5.1	3.4	1.5	0.2	<i>I. setosa</i>
5	3.5	1.3	0.3	<i>I. setosa</i>
4.5	2.3	1.3	0.3	<i>I. setosa</i>
4.4	3.2	1.3	0.2	<i>I. setosa</i>
5	3.5	1.6	0.6	<i>I. setosa</i>
5.1	3.8	1.9	0.4	<i>I. setosa</i>
4.8	3	1.4	0.3	<i>I. setosa</i>
5.1	3.8	1.6	0.2	<i>I. setosa</i>
4.6	3.2	1.4	0.2	<i>I. setosa</i>
5.3	3.7	1.5	0.2	<i>I. setosa</i>
5	3.3	1.4	0.2	<i>I. setosa</i>
7	3.2	4.7	1.4	<i>I. versicolor</i>
6.4	3.2	4.5	1.5	<i>I. versicolor</i>
6.9	3.1	4.9	1.5	<i>I. versicolor</i>
5.5	2.3	4	1.3	<i>I. versicolor</i>
6.5	2.8	4.6	1.5	<i>I. versicolor</i>
5.7	2.8	4.5	1.3	<i>I. versicolor</i>
6.3	3.3	4.7	1.6	<i>I. versicolor</i>
4.9	2.4	3.3	1	<i>I. versicolor</i>
6.6	2.9	4.6	1.3	<i>I. versicolor</i>
5.2	2.7	3.9	1.4	<i>I. versicolor</i>
5	2	3.5	1	<i>I. versicolor</i>
5.9	3	4.2	1.5	<i>I. versicolor</i>
6	2.2	4	1	<i>I. versicolor</i>
6.1	2.9	4.7	1.4	<i>I. versicolor</i>
5.6	2.9	3.6	1.3	<i>I. versicolor</i>
6.7	3.1	4.4	1.4	<i>I. versicolor</i>
5.6	3	4.5	1.5	<i>I. versicolor</i>
5.8	2.7	4.1	1	<i>I. versicolor</i>
6.2	2.2	4.5	1.5	<i>I. versicolor</i>
5.6	2.5	3.9	1.1	<i>I. versicolor</i>
5.9	3.2	4.8	1.8	<i>I. versicolor</i>

Sepal length	Sepal width	Petal length	Petal width	Species
6.1	2.8	4	1.3	<i>I. versicolor</i>
6.3	2.5	4.9	1.5	<i>I. versicolor</i>
6.1	2.8	4.7	1.2	<i>I. versicolor</i>
6.4	2.9	4.3	1.3	<i>I. versicolor</i>
6.6	3	4.4	1.4	<i>I. versicolor</i>
6.8	2.8	4.8	1.4	<i>I. versicolor</i>
6.7	3	5	1.7	<i>I. versicolor</i>
6	2.9	4.5	1.5	<i>I. versicolor</i>
5.7	2.6	3.5	1	<i>I. versicolor</i>
5.5	2.4	3.8	1.1	<i>I. versicolor</i>
5.5	2.4	3.7	1	<i>I. versicolor</i>
5.8	2.7	3.9	1.2	<i>I. versicolor</i>
6	2.7	5.1	1.6	<i>I. versicolor</i>
5.4	3	4.5	1.5	<i>I. versicolor</i>
6	3.4	4.5	1.6	<i>I. versicolor</i>
6.7	3.1	4.7	1.5	<i>I. versicolor</i>
6.3	2.3	4.4	1.3	<i>I. versicolor</i>
5.6	3	4.1	1.3	<i>I. versicolor</i>
5.5	2.5	4	1.3	<i>I. versicolor</i>
5.5	2.6	4.4	1.2	<i>I. versicolor</i>
6.1	3	4.6	1.4	<i>I. versicolor</i>
5.8	2.6	4	1.2	<i>I. versicolor</i>
5	2.3	3.3	1	<i>I. versicolor</i>
5.6	2.7	4.2	1.3	<i>I. versicolor</i>
5.7	3	4.2	1.2	<i>I. versicolor</i>
5.7	2.9	4.2	1.3	<i>I. versicolor</i>
6.2	2.9	4.3	1.3	<i>I. versicolor</i>
5.1	2.5	3	1.1	<i>I. versicolor</i>
5.7	2.8	4.1	1.3	<i>I. versicolor</i>
6.3	3.3	6	2.5	<i>I. virginica</i>
5.8	2.7	5.1	1.9	<i>I. virginica</i>
7.1	3	5.9	2.1	<i>I. virginica</i>
6.3	2.9	5.6	1.8	<i>I. virginica</i>
6.5	3	5.8	2.2	<i>I. virginica</i>
7.6	3	6.6	2.1	<i>I. virginica</i>
4.9	2.5	4.5	1.7	<i>I. virginica</i>
7.3	2.9	6.3	1.8	<i>I. virginica</i>

Sepal length	Sepal width	Petal length	Petal width	Species
6.7	2.5	5.8	1.8	<i>I. virginica</i>
7.2	3.6	6.1	2.5	<i>I. virginica</i>
6.5	3.2	5.1	2	<i>I. virginica</i>
6.4	2.7	5.3	1.9	<i>I. virginica</i>
6.8	3	5.5	2.1	<i>I. virginica</i>
5.7	2.5	5	2	<i>I. virginica</i>
5.8	2.8	5.1	2.4	<i>I. virginica</i>
6.4	3.2	5.3	2.3	<i>I. virginica</i>
6.5	3	5.5	1.8	<i>I. virginica</i>
7.7	3.8	6.7	2.2	<i>I. virginica</i>
7.7	2.6	6.9	2.3	<i>I. virginica</i>
6	2.2	5	1.5	<i>I. virginica</i>
6.9	3.2	5.7	2.3	<i>I. virginica</i>
5.6	2.8	4.9	2	<i>I. virginica</i>
7.7	2.8	6.7	2	<i>I. virginica</i>
6.3	2.7	4.9	1.8	<i>I. virginica</i>
6.7	3.3	5.7	2.1	<i>I. virginica</i>
7.2	3.2	6	1.8	<i>I. virginica</i>
6.2	2.8	4.8	1.8	<i>I. virginica</i>
6.1	3	4.9	1.8	<i>I. virginica</i>
6.4	2.8	5.6	2.1	<i>I. virginica</i>
7.2	3	5.8	1.6	<i>I. virginica</i>
7.4	2.8	6.1	1.9	<i>I. virginica</i>
7.9	3.8	6.4	2	<i>I. virginica</i>
6.4	2.8	5.6	2.2	<i>I. virginica</i>
6.3	2.8	5.1	1.5	<i>I. virginica</i>
6.1	2.6	5.6	1.4	<i>I. virginica</i>
7.7	3	6.1	2.3	<i>I. virginica</i>
6.3	3.4	5.6	2.4	<i>I. virginica</i>
6.4	3.1	5.5	1.8	<i>I. virginica</i>
6	3	4.8	1.8	<i>I. virginica</i>
6.9	3.1	5.4	2.1	<i>I. virginica</i>
6.7	3.1	5.6	2.4	<i>I. virginica</i>
6.9	3.1	5.1	2.3	<i>I. virginica</i>
5.8	2.7	5.1	1.9	<i>I. virginica</i>
6.8	3.2	5.9	2.3	<i>I. virginica</i>
6.7	3.3	5.7	2.5	<i>I. virginica</i>

Sepal length	Sepal width	Petal length	Petal width	Species
6.7	3	5.2	2.3	<i>I. virginica</i>
6.3	2.5	5	1.9	<i>I. virginica</i>
6.5	3	5.2	2	<i>I. virginica</i>
6.2	3.4	5.4	2.3	<i>I. virginica</i>
5.9	3	5.1	1.8	<i>I. virginica</i>

APPENDIX B

Curriculum Vitae # Example

PERSONAL INFORMATION

NAME : Mr. Handsome
 DATE AND PLACE OF BIRTH : May 04, 2015 - Bandung, West Java
 ADDRESS : ...
 CITY AND PROVINCE : ...
 POSTAL CODE : 99999
 SEX : Male
 NATIONALITY : Indonesia
 RELIGION : ...
 MARITAL STATUS : Single
 LANGUAGE SKILL 1 : Indonesian (Mother Tongue)
 LANGUAGE SKILL 2 : English (Sufficient)
 LANGUAGE SKILL 3 : Minang (Sufficient)
 LANGUAGE SKILL 4 : Java (Intermediate)
 LANGUAGE SKILL 5 : Malay (Basic Knowledge)
 LANGUAGE SKILL 6 : Arabic (Basic Knowledge)
 PHONE 1 : +62 851 xxxx xxxx
 PHONE 2 : +62 852 xxxx xxxx
 FAX : -
 EMAIL : youremail@students.telkomuniversity.ac.id

EDUCATION

Education Degree	Institution	Period of Education	Year of Graduation
Master	Telkom University	2013 - 2015	2015
Bachelor	...	2008 - 2012	2012
High School	...	2007 - 2008	2008
Middle School	...	2003 - 2005	2005

PUBLICATIONS

Title	Event	Year
...	...	2011
...	...	2012

TEACHING AND PROFESSIONAL EXPERIENCE

Position	Institution	Field	Period
...	...	...	2012 - 2013
...	...	...	2009 - 2013

TRAINING AND WORKSHOP

Training and Workshop	Organizer	Year
...	...	2012
...	...	2009

ACHIEVEMENT

Achievement	Event	Year
1 th	Winner of The Awesome Thesis 2015, Telkom University	2015